DEEP LEARNING · BREAST HISTOPATHOLOGY

BreakHis: benign vs malignant

Patient-level cross-validation, MLflow experiment tracking, and traceable predictions with fastai and PyTorch

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THE QUESTION

Can a CNN tell benign from malignant breast tissue in patients it has never seen?

Input

H&E microscopy image of breast tissue at 200X magnification

Output

Probability the tissue is malignant, per image and averaged per patient

Standard

Evaluate on unseen patients, and log every run so any prediction can be traced

THE DATA

BreakHis: 7,909 images from 82 patients

2,480
benign images

5,429
malignant images (about 2× more)

4
magnifications: 40X, 100X, 200X, 400X

Class	Tumor types
Benign	Adenosis, fibroadenoma, phyllodes tumor, tubular adenoma
Malignant	Ductal, lobular, mucinous, papillary carcinoma

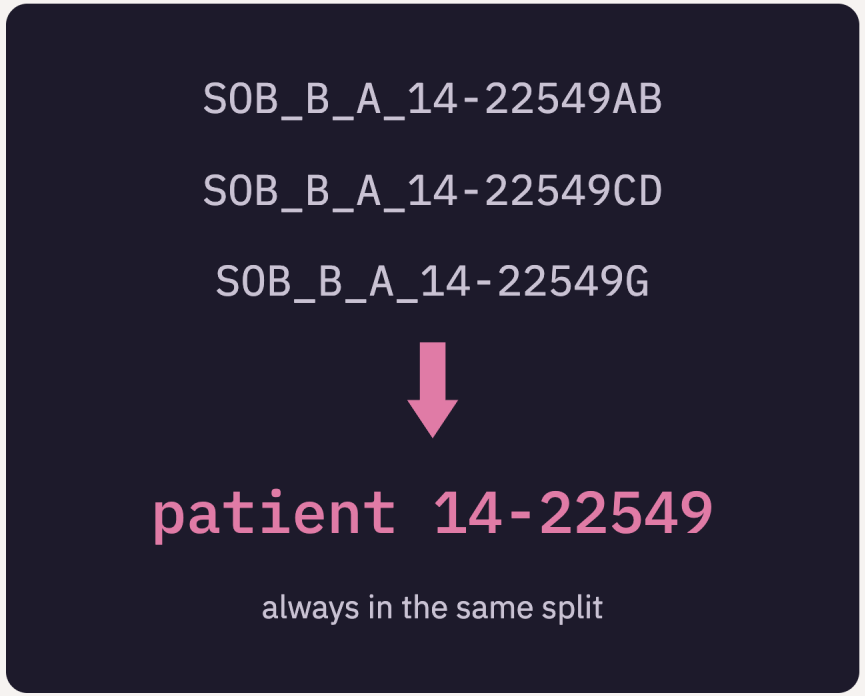
This project: 200X → 2,013 images, 70 patients

METHOD · AVOIDING DATA LEAKAGE

Split by patient, not by image

A network can memorize a patient's staining and slide preparation. If one patient is in both training and validation, the model is rewarded for recognizing the patient, not the disease.

Some folders share a patient number with different letter suffixes, likely separate slides from the same patient. I grouped them by number so each patient stays on one side of the split.



PIPELINE

From image to tracked prediction

Images
200X PNGs; patient and tumor type parsed from filename

→

DataBlock
Resize to 224 px; flips, rotation, zoom

→

ResNet34
ImageNet weights, fastai fine_tune

→

Predictions
p(malignant) per image, averaged per patient

→

Evaluation
AUC, confusion matrix, top losses

MLflow logs every step

Parameters · per-epoch metrics · validation patient list · figures · predictions · features · model

fastai · PyTorch · scikit-learn · pandas · MLflow (SQLite backend) · Kaggle T4 GPU · 5

RESULT 1 · ONE PATIENT-LEVEL SPLIT

One split looked weak, and benign tissue failed most

0.76

image-level AUC

0.82

patient-level AUC

Malignant: 242 of 279 correct (87%)



Benign: 45 of 111 correct (41%)



Validation loss rose after epoch 1 while training loss kept falling: a sign of overfitting.

ResNet34, 4 epochs, ~20% of patients held out per class · 6

ERROR ANALYSIS

Errors traced back to individual patients

0 / 32

fibroadenoma images correct, all from one patient (14-25197)

45% · 90%

accuracy for the two tubular adenoma patients

3

benign patients in the whole validation set; three tumor types absent

→ **One split isn't representative, so test every patient with cross-validation**

Accuracy per tumor type and per patient, threshold 0.5 · 7

RESULT 2 · 5-FOLD PATIENT-LEVEL CROSS-VALIDATION

Every patient tested once

0.954

out-of-fold patient AUC across all 70 patients

Evaluation	Image AUC	Patient AUC
Single split	0.758	0.818
CV, 2 epochs	0.895 ± 0.077	0.959 ± 0.068
CV, 4 epochs	0.895 ± 0.074	0.962 ± 0.038

Evaluation design mattered more than training length.

StratifiedGroupKFold: groups = patient, stratified by tumor type · final model: 4 epochs · 8

OUT-OF-FOLD ACCURACY BY TUMOR TYPE

Common tumor types are classified best

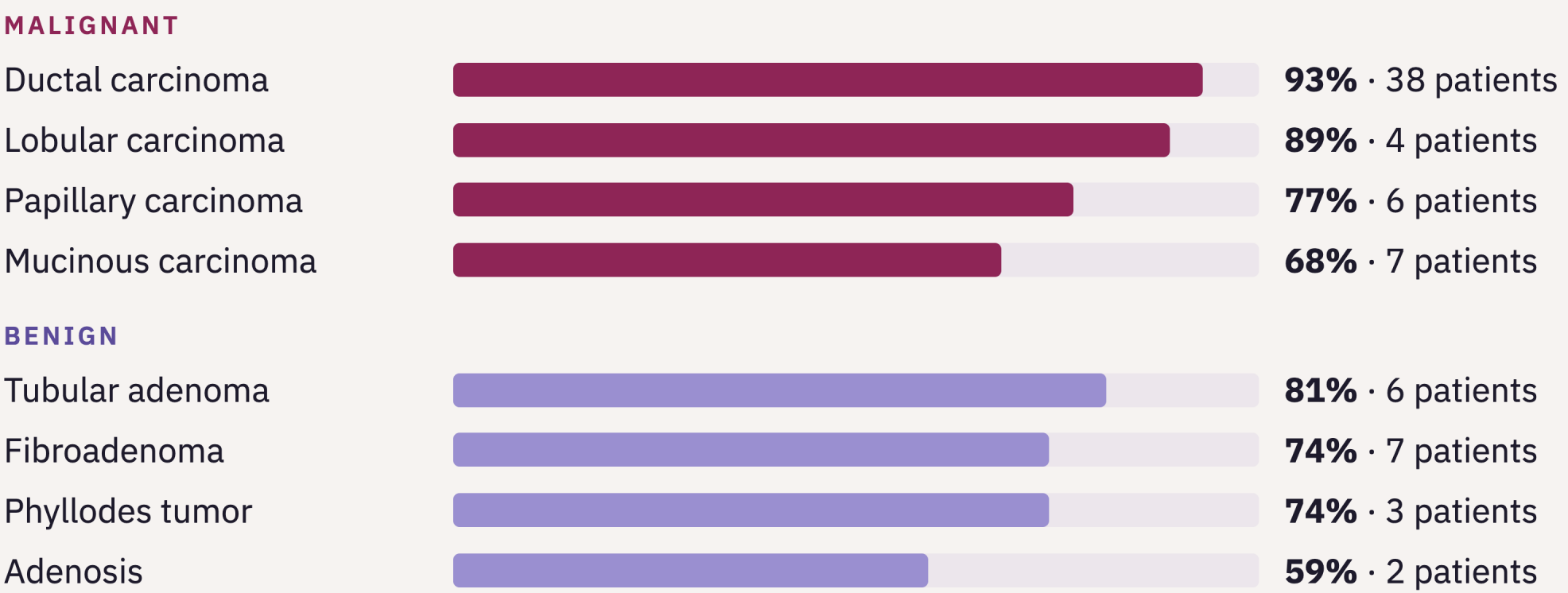


Image accuracy at threshold 0.5, 2-epoch CV run · fibroadenoma errors come from 2 of 7 patients · 9

TRACEABILITY · MLFLOW

Every run is tracked in MLflow

Parameters

Architecture, epochs, image size, batch size, magnification, seed, split method

Metrics

Loss and AUC per epoch, AUC per fold, CV mean ± std, out-of-fold patient AUC

Artifacts

Validation patient list, confusion matrix, top-loss images, out-of-fold predictions

Final model run

Trained model, 512-feature table per image, PCA plot, link to the CV run that chose it

TRACEABILITY DEMO · FEATURE EXPORT

Tracing the most confident error

SOB_M_PC-14-9146-200-011.png

Field	Value
Tumor type	Papillary (M)
Patient	14-9146
CV p(malignant)	0.00
Final p(malignant)	0.47

Nearest neighbors in feature space, other patients

- 3 fibroadenoma, benign (patient 14-21998)
- 1 phyllodes tumor, benign
- 2 papillary carcinoma, malignant

Its papillary structure resembles duct-rich benign lesions, and papillary carcinoma is rare here (6 patients).

Cosine similarity on 512-dim features · same-patient images excluded · trace logged to MLflow · 11

WRAP-UP

Limitations and next steps

Limitations

- One magnification (200X), one lab
- 70 patients, only about 18 benign
- No stain normalization; threshold fixed at 0.5
- No external validation

Next steps

- Compare 40X, 100X, 400X with run_cv
- Stain normalization and tumor-type weighting
- Validate on an external dataset
- Apply the pipeline to mammograms (CBIS-DDSM)

Patient-level evaluation and traceable runs carry over directly to clinical imaging.